CDA 106
Numerical Methods for Data Science

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Result:

Any $n \times n$ matrix A has LU decomposition if and only if all the leading principle minors are non-zero.

Principle minors

$$A = \begin{bmatrix} \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \end{bmatrix}$$

$A = LU$
$n \times n$

$n^2 + \cancel{n}$ eq in $n^2 + n$ var.

fix

$Ax = b$

$L(Ux) = b$

$Ly = b$, $Ux = y$

Result: If Matrix A is 'positive definite' then it has LU decomposion.

A is positive definite $\Rightarrow$ A has LU decomposition

For A to have LU decomposition, being 'positive definite' is sufficient cond^n.

If and only if (Necessary & sufficient)

Statement A $\Rightarrow$ Statement B

then A is sufficient for B

Statement B is considered as necessary.

And statement B $\Rightarrow$ Statement A.

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eg. A is invertible $\Rightarrow |A| \neq 0$ ✓

$\Rightarrow$ A full rank. ✓ ✓

f is Diff $\Rightarrow$ f is Cont.

① Diff suff. Cont.

② Cont necessary Diff.

Cont is necessary but not suff. for diff.

st A $\Rightarrow$ st B.

& st B $\Rightarrow$ st A

st A is both necessary & sufficient for B.

"if and only if"

LU decomposition is also very useful in finding inverse of a matrix.

$A_{n\times n} \longrightarrow A^{-1}$ finding is little complex

<u>Idea</u>

$AX = b \Rightarrow X = A^{-1}b \quad (|A| \neq 0)$

$A = LU \Rightarrow LUX = b$

$\Rightarrow UX = L^{-1}b$

$\Rightarrow X = U^{-1}L^{-1}b$

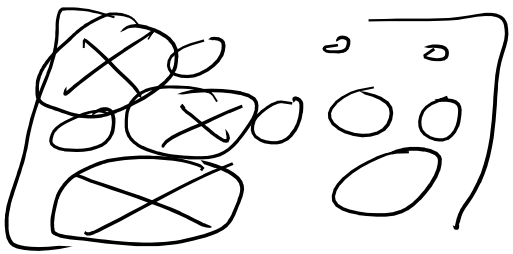
$\boxed{A^{-1} = U^{-1}L^{-1}}$

$L = \begin{bmatrix} \times & 0 \\ \times & \end{bmatrix}$ & $U = \begin{bmatrix} & \\ 0 & \end{bmatrix}$

Min Zeros of L or $U = n^2 - \left\{\frac{n^2+n}{2}\right\} = \frac{n^2-n}{2}$

Because of so many zeros in L & U, calculating L^{-1} & U^{-1} is far more easier than calculating A^{-1} directly

Sparse when there are a lot of zeros in a matrix then the matrix is referred as <u>sparse</u> matrix.



Consider $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ $\quad |0| = 0 \quad \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = 1$

A does not have LU decomposition.

Que can we find L & U matrices for above A

I will try and address this question.

$$A_1 = \begin{bmatrix} 3 & 2 & -3 \\ 2 & 2 & 5 \\ 1 & 1 & -1 \end{bmatrix}$$

$|a_{11}| = |3| = 3$

$\begin{vmatrix} 3 & 2 \\ 2 & 2 \end{vmatrix} = 2$ $\quad |A| = -7$

~~ABSO~~ Also

$$A_2 = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 2 & 5 \\ 3 & 2 & -3 \end{bmatrix}$$

$|a_{11}| = |1| = 1$, $\begin{vmatrix} 1 & 1 \\ 2 & 2 \end{vmatrix} = 0$

$|A| = 7$

$A_1 \sim A_2$ (Row exchange)

Our aim is to find P, L, & U for any A:

$$\underline{\underline{PA}} = \underline{\underline{LU}}$$

This is referred as PLU decomposition.

Permutation Matrix

Set $\underbrace{S = \{a, b, c\}}$

$$\left.\begin{aligned} S_1 &= (a, b, c) \\ S_2 &= (a, c, b) \\ S_3 &= (b, a, c) \\ S_4 &= (b, c, a) \\ S_5 &= (c, a, b) \\ S_6 &= (c, b, a) \end{aligned}\right\} \quad 6 \text{ permutations}$$

$$\begin{array}{ccc} |\times & |\times & |\times \\ \uparrow & \uparrow & \uparrow \\ 3 & 2 & 1 \end{array} = 3 \times 2 \times 1 = \underline{\underline{6}}$$

Permutation Matrix of I_3

$$I_3 = \begin{array}{c} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{array} \begin{array}{l} \rightarrow R_1 \\ \rightarrow R_2 \\ \rightarrow R_3 \end{array}$$

$$P_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3$$

$$P_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{array}{l} R_1 \\ R_3 \\ R_2 \end{array}$$

$$P_3 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{array}{l} R_2 \\ R_1 \\ R_3 \end{array}$$

$$P_4 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{array}{l} R_2 \\ R_3 \\ R_1 \end{array}$$

6 permutation Matrices

$$(\underbrace{R_1}_{a}, \underbrace{R_2}_{b}, \underbrace{R_3}_{c})$$

$$P_5 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{array}{l} R_3 \\ R_1 \\ R_2 \end{array}$$

$$P_6 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{array}{l} R_3 \\ R_2 \\ R_1 \end{array}$$

$P_1 - P_6$ are our 6 permutations of I_3.

Elementary Matrices

Elementary matrices are obtained from Identity matrices by <u>single</u> elementary (row) operations.

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Elementary row operations

1. Row exchange $i \leftrightarrow j$
2. ith row $\times k$
3. ith row $\times k$ and add to jth row

Example

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

① $I_3 \xrightarrow[R_1 \leftrightarrow R_2]{} E_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ row exchange

Permutation matrix

② $I_3 \xrightarrow[R_3(k)]{} E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & k \end{bmatrix}$

③ $I_3 \xrightarrow[R_1 + R_3(k)]{} E_3 \quad \begin{bmatrix} 1 & 0 & k \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

E_1, E_2 and E_3 are examples of elementary matrices.

Now we shall see the effect of premultiplication of an elementary matrix.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$\underbrace{E_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}}_{\text{elementary matrix}} \longleftarrow R_1 \leftrightarrow R_2 \qquad I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Now, we pre-multiply E_1 to A

$$E_1 A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$$

$R_1 \leftrightarrow R_2$ in Matrix A

- Therefore, the pre-multiplication of elementary matrix E_1 to A gives the exchange of row 1 to row 2 of A.
- Therefore, if I want to multiply 2nd row of matrix A by 2, then, I should premultiply A by an elementary matrix E_2 where E_2 is obtained by multiplying 2nd row of I_2 by 2. i.e. $E_2 = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

$$\Rightarrow \quad E_2 A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 6 & 8 \end{bmatrix}$$

Similarly, suppose you want 2 time 2nd row Added to the first row of Matrix A, then ~~pre~~ premultiplication with

$$E_3 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

$(R_1 \to R_1 + 2R_2$ in $I_2)$

$$E_3 A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 7 & 10 \\ 3 & 4 \end{bmatrix}$$

Therefore, if we want to perform a row operation on a matrix, we can do so by premultiplying it with a suitable elementary matrix.

Now consider $A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$$

$E_1 A$

$E_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

$E_2(E_1 A)$

$R_2 - R_2(2)$

$$\begin{bmatrix} 1 & 4 \\ 0 & -5 \end{bmatrix}$$

$E_2 = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$

$R_2(-\frac{1}{5})$

$$\underbrace{\begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix}}_{\text{RREF}}$$

$E_3 = \begin{bmatrix} 1 & 0 \\ 0 & -1/5 \end{bmatrix}$

$E_3(E_2 E_1 A)$

$$\Rightarrow E_3 E_2 E_1 A = \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix} = U$$

RREF

Row reduced Echelon form

$$\begin{bmatrix} 1 & x & x & x & x \\ 0 & 1 & x & x & x \\ 0 & 0 & 1 & x & x \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$E_3 E_2 E_1 A = U$$

$$\Rightarrow E_2 E_1 A = E_3^{-1} U$$

$$\Rightarrow A = \underbrace{E_1^{-1} E_2^{-1} E_3^{-1}}_{L} U$$

Elementary matrices are invertible (non-singular)

expectation

so that $\underline{A = LU}$

Now let us consider $A = \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix}$

we cannot find LU decomposition.

$$A = \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix} \xrightarrow{E_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}} \begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix} \xrightarrow{E_2 = \begin{bmatrix} 1/2 & 0 \\ 0 & 1 \end{bmatrix}} \begin{bmatrix} 1 & 3/2 \\ 0 & 1 \end{bmatrix} = U$$

RREF

$$E_2 E_1 A = U$$

$$\Rightarrow A = \underbrace{E_1^{-1} E_2^{-1}}_{?} U$$

Check whether $E_1^{-1} E_2^{-1}$ is a lower triangular matrix?

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$|A| = ad - bc$$

$$\text{Cofactor } A = \begin{bmatrix} d & -c \\ -b & a \end{bmatrix}$$

$$\text{Adj } A = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$X \longrightarrow X \longrightarrow X$$